

1. Research Question

Core question: Does pollination success depend only on how many pollinators visit, or also on who visits and when?

- Tests whether community-weighted mean (CWM) bee tongue length, CWM bee body size, and morning visit fraction predict garden-level seed/fruit set.
- Asks whether trait effects are stronger for plants with specialised flower morphologies such as comfrey and sainfoin.
- Research value: urban densification changes pollinator communities; better metrics can identify which gardens provide effective pollination service.



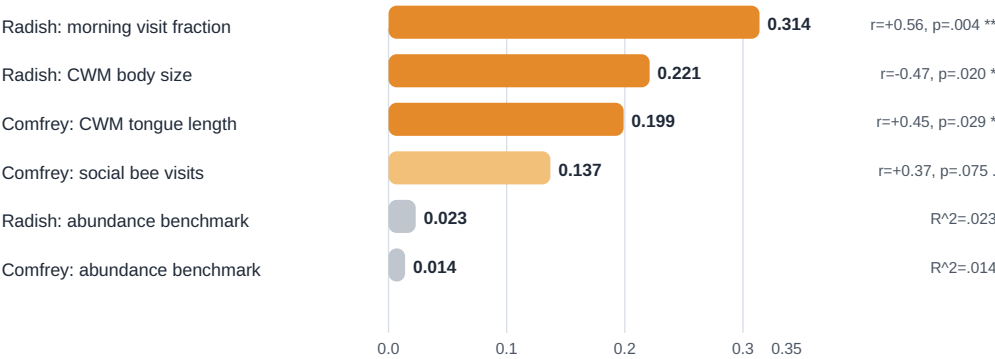
2. Methodology / Project Design



- Data source: open Zurich home-garden experiment from Reji Chacko, Moretti & Frey (2025), Data in Brief / EnviDat.
- Pollination outcomes: carrot seeds per umbel; radish, sainfoin, and comfrey fruit-set proportions.
- CWM trait cleaning: zero trait values treated as missing; nectar robbers excluded from legitimate-pollinator CWM calculations.
- Temporal variables: visits grouped into morning (09:00-12:00), midday (12:00-15:00), and afternoon (15:00-19:00); evenness = Shannon H / log(n).
- Tools: Python 3.10, pandas, NumPy, SciPy, Matplotlib; Claude Opus and ChatGPT 5.5 assisted cleaning, analysis, and graph visualization.

3. Data Analysis & Results

Variance explained (R²) by selected predictors n = 24 gardens per plant; exploratory correlations



Strongest: radish morning timing (R²=.314) Trait matching: comfrey tongue length (R²=.199)

4. Interpretation & Conclusions

Answer: Pollinator identity and visit timing added predictive information beyond visit count, but effects were plant-specific.

- Comfrey: longer-tongued bee communities achieved higher fruit set, matching its deeply concealed nectar morphology.
- Radish: morning-concentrated visitation was the strongest predictor, possibly aligning with peak floral receptivity.
- Carrot: no significant trait or temporal predictor, consistent with an open generalist flower where many visitors can access nectar.
- Innovation: reframes the dataset from abundance response to trait- and time-based predictors of pollination service.
- Limitations: n = 24 gardens; exploratory correlations; no landscape covariates; no direct pollen deposition or receptivity measurements.
- Future work: mixed-effects models, functional diversity metrics, replication across Swiss cities, and single-visit pollen deposition tests.